

Inequalities – 1

The Cauchy–Bunyakovsky inequality as a technique

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Theoretical minimum.

Theorem 1 (three forms of one inequality). Discrete: $(\sum a_i b_i)^2 \leq \sum a_i^2 \sum b_i^2$.

Integral: $(\int f g)^2 \leq \int f^2 \int g^2$. In an inner product space: $|\langle f, g \rangle| \leq \|f\| \|g\|$.

All three are the statement that the quadratic $\lambda \mapsto \int (f - \lambda g)^2 \geq 0$ has non-positive discriminant, and equality holds exactly when f and g are proportional.

Theorem 2 (the Engel form). For $b_i > 0$ one has $\sum \frac{a_i^2}{b_i} \geq \frac{(\sum a_i)^2}{\sum b_i}$, and in the integral form $\int \frac{f^2}{g} \geq \frac{(\int f)^2}{\int g}$ for $g > 0$. This is the shape in which the inequality is actually applied: it converts a sum of fractions into one fraction.

Fact 1 (choosing the weights). The whole art is in splitting the integrand: to bound $\int h$ one writes $h = f \cdot g$ so that $\int f^2$ and $\int g^2$ are computable, and the equality case tells which splitting is right. If the answer of a problem is attained at some f_0 , then the correct splitting is the one making f proportional to g at f_0 .

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1. Prove all three forms of Theorem 1 through the discriminant of $\int (f - \lambda g)^2$, describe the equality case, and derive the Engel form of Theorem 2 from the discrete inequality.
 2. (RMO 2004) Let $f: [0, 1] \rightarrow \mathbb{R}$ be integrable with $\int_0^1 f = \int_0^1 x f = 1$. Prove that $\int_0^1 f^2 \geq 4$.
 3. (RMO) Let n be a positive integer and let $f: [0, 1] \rightarrow \mathbb{R}$ be continuous with $\int_0^1 x^k f(x) dx = 1$ for every $k \in \{0, 1, \dots, n-1\}$. Prove that $\int_0^1 f^2 \geq n^2$, and explain how the previous problem is the case $n = 2$.

4. (VJIMC 2002) Positive numbers $x_1, \dots, x_n$ satisfy $\sum_{i=1}^n \frac{1}{1+x_i} = 1$.
Prove that $\sum \sqrt{x_i} \geq (n-1) \sum \frac{1}{\sqrt{x_i}}$.
5. (IMO 1995) Let a, b, c be positive reals with $abc = 1$. Prove that $\frac{1}{a^3(b+c)} + \frac{1}{b^3(c+a)} + \frac{1}{c^3(a+b)} \geq \frac{3}{2}$, and point out where the Engel form does the work. (Attention: AM-GM inequality is banned! Prove this inequality using only Cauchy-Schwarz and Engel)